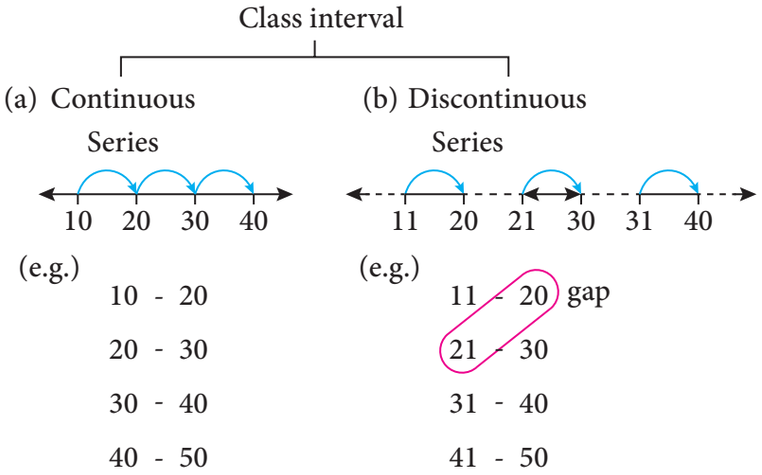
**For example,**

Marks for the C.I 10 to 20 can be written as 10-20, whose class size is 20–10=10




(a) While distributing the frequency, we follow the counting as given below. Suppose the classes are 10-20, 20-30, 30-40, 40-50 ..... This represent a continuous series. Here, 20 is included in the class 20-30 and 30 is included in 30-40, likewise for the other classes also.

(b) In case the given series has a gap between the limits of any two adjacent classes, this gap may be filled up by extending the two limits of each class by taking half of the value of the gap. Half of the gap is called the adjustment factor.

**Conversion of a discontinuous series into continuous series:**

In case the given series is a discontinuous, we can make it as continuous as follows,

**Illustration 1:**

11 - 20 gap
21 - 30
31 - 40
41 - 50

difference in the gap = 21 – 20
= 1

Lower boundary = lower limit –half of the gap

$$= 11 - \frac{1}{2}(1)$$

= 11 – 0.5 = 10.5

Upper boundary = upper limit + half of the gap

$$= 20 + \frac{1}{2}(1)$$

= 20 + 0.5

= 20.5 and so on for other classes too.

Therefore, the class interval can be changed into a continuous one as given in the following table,

| Discontinuous series | Continuous series |
|---|---|
| 11-20 (-0.5, +0.5) | 10.5-20.5 |
| 21-30 | 20.5-30.5 |
| 31-40 | 30.5-40.5 |
| 41-50 | 40.5-50.5 |



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